Light

GCSE Computer Science – Computational Thinking (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Why is decomposition useful when solving complex problems?

- ○ **A.** It allows different people to work on different parts of the problem simultaneously
- ○ **B.** It makes problems more complicated
- ○ **C.** It removes the need for testing
- ○ **D.** It automatically writes the code for you

Q2 (1 mark)

A map of the London Underground is an example of abstraction because it:

- ○ **A.** Removes unnecessary geographical detail to focus on stations and connections
- ○ **B.** Shows every street in London in detail
- ○ **C.** Is printed on paper
- ○ **D.** Uses the same scale as a road map

Q3 (1 mark)

In a flowchart, what does an oval (rounded rectangle) represent?

- ○ **A.** A process or calculation
- ○ **B.** A decision point
- ○ **C.** An input or output operation
- ○ **D.** The start or end of the algorithm (terminator)

Q4 (1 mark)

Which of the following is NOT a characteristic of a well-designed algorithm?

- ○ **A.** It has a clear sequence of steps
- ○ **B.** It produces a defined output
- ○ **C.** It runs indefinitely without stopping
- ○ **D.** It can be followed without ambiguity

Q5 (1 mark)

In a ticket vending machine, what would be classified as an output?

- ○ **A.** The buttons pressed by the customer
- ○ **B.** The money inserted into the machine
- ○ **C.** The destination selected on screen
- ○ **D.** The printed ticket dispensed to the customer

Q6 (1 mark)

How does pattern recognition help programmers?

- ○ **A.** It helps them recognise faces in photographs
- ○ **B.** It automatically fixes bugs in code
- ○ **C.** It teaches them to type faster
- ○ **D.** It allows them to identify similarities in problems so they can reuse existing solutions

Q7 (1 mark)

In pseudocode, what keyword is typically used to repeat a block of code while a condition remains true?

- ○ **A.** WHILE
- ○ **B.** IF
- ○ **C.** OUTPUT

- ○ **D.** FUNCTION

Q8 (1 mark)

In a flowchart, what shape represents a process or calculation?

- ○ **A.** Diamond
- ○ **B.** Oval
- ○ **C.** Parallelogram
- ○ **D.** Rectangle

Q9 (1 mark)

In a structure diagram, if 'Make Tea' is at the top level with 'Boil Water', 'Add Tea Bag', and 'Add Milk' below it, what does this show?

- ○ **A.** 'Make Tea' has been decomposed into three sub-tasks
- ○ **B.** The three tasks must be done in that exact order
- ○ **C.** Only one of the three tasks needs to be completed
- ○ **D.** The tasks are unrelated to making tea

Q10 (1 mark)

What are the four main components (pillars) of computational thinking?

- ○ **A.** Hardware, Software, Networks, Security

- ○ **B.** Input, Process, Output, Storage
- ○ **C.** Decomposition, Pattern Recognition, Abstraction, Algorithmic Thinking
- ○ **D.** Variables, Loops, Functions, Arrays

Q11 (1 mark)

What does the MOD operator return?

- ○ **A.** The result of multiplying two numbers
- ○ **B.** The remainder after integer division
- ○ **C.** The average of two numbers
- ○ **D.** The larger of two numbers

Q12 (1 mark)

What best describes an algorithm?

- ○ **A.** A finite sequence of well-defined instructions to solve a problem or perform a task
- ○ **B.** A type of computer hardware
- ○ **C.** A programming language
- ○ **D.** A type of computer virus

Q13 (2 marks)

A school wants to create a system to manage student attendance. Using decomposition, which breakdown would be most appropriate?

- ○ **A.** Morning, Afternoon, Evening
- ○ **B.** Year 7, Year 8, Year 9, Year 10, Year 11
- ○ **C.** Monday, Tuesday, Wednesday, Thursday, Friday
- ○ **D.** Record attendance, Generate reports, Send notifications to parents, Manage student records

Q14 (2 marks)

When designing a car racing video game, a developer decides to model cars with speed, acceleration, and handling attributes, but ignores engine valve timing and fuel injection rates.
What level of abstraction is being applied?

- ○ **A.** No abstraction — all details are included
- ○ **B.** Too much abstraction — essential details are missing
- ○ **C.** Appropriate abstraction — relevant gameplay factors are included while unnecessary mechanical details are hidden
- ○ **D.** The developer has made an error and should include all engine details

Q15 (2 marks)

A flowchart shows: Start → Set count to 0 → Is count < 3? → (Yes) Output "Hello" → Add 1 to count → [arrow back to decision] → (No) End
How many times will "Hello" be output?

- A. 0 times
- B. 2 times
- C. 3 times
- D. Infinite times

Q16 (2 marks)

Consider the following pseudocode:

```
x = 5
y = 2
WHILE x > 0
    y = y + x
    x = x - 1
ENDWHILE
OUTPUT y
```

What value will be output?

- A. 2
- B. 15
- C. 17
- D. 20

Q17 (2 marks)

A fitness tracker monitors heart rate, counts steps, and displays daily activity summaries.
Identify which of these is a process.

- ○ **A.** The heart rate sensor reading pulse data
- ○ **B.** Calculating the total number of calories burned based on steps and heart rate
- ○ **C.** The screen displaying the daily summary
- ○ **D.** The strap holding the tracker on the wrist

Q18 (2 marks)

A programmer has written code to validate that an email address contains an @ symbol. They later need to validate that a Twitter handle contains an @ symbol.
What should pattern recognition lead them to do?

- ○ **A.** Recognise the common pattern and create a generalised 'contains character' function that can check for any symbol in any string
- ○ **B.** Write completely new code from scratch for Twitter handles
- ○ **C.** Delete the email validation code
- ○ **D.** Combine email and Twitter validation into one function that only works for these two cases

Q19 (2 marks)

In a flowchart, a diamond shape has two arrows leaving it labelled 'Yes' and 'No'. The 'Yes' arrow leads to 'Apply discount' and the 'No' arrow leads to 'Calculate full price'. Both then converge to 'Print receipt'.
What programming construct does this represent?

- ○ **A.** A FOR loop that repeats the discount calculation
- ○ **B.** A WHILE loop that keeps applying discounts

- ○ **C.** Selection (IF-ELSE) where different actions occur based on a condition
- ○ **D.** Sequence where all steps happen one after another

Q20 (2 marks)

Consider the following pseudocode:

```
num = 47
tens = num DIV 10
units = num MOD 10
OUTPUT tens
OUTPUT units
```

What values will be output?

- ○ **A.** tens = 47, units = 0
- ○ **B.** tens = 7, units = 4
- ○ **C.** tens = 4.7, units = 0.7
- ○ **D.** tens = 4, units = 7

Q21 (2 marks)

A structure diagram for a 'Calculator' application shows four boxes at the second level: 'Addition', 'Subtraction', 'Multiplication', 'Division'. Each of these then has 'Get Numbers' and 'Display Result' below them.
What problem does this structure have?

- ○ **A.** The diagram is perfect and has no problems
- ○ **B.** There are too many levels in the hierarchy

- ○ **C.** 'Get Numbers' and 'Display Result' are duplicated under each operation when they could be shared common components
- ○ **D.** Calculators cannot perform these operations

Q22 (2 marks)

A student writes an algorithm to make a cup of instant coffee: 1) Boil water 2) Pour water into cup 3) Add coffee granules 4) Stir
What is wrong with this algorithm?

- ○ **A.** Nothing is wrong — the steps are in the correct order
- ○ **B.** The coffee granules should be added before the water is poured, otherwise they won't dissolve properly
- ○ **C.** Stirring is not necessary
- ○ **D.** The algorithm should include switching on the television

Q23 (2 marks)

Consider the following pseudocode:

```
age = 25
IF age >= 18 THEN
    IF age >= 65 THEN
        OUTPUT "Senior"
    ELSE
        OUTPUT "Adult"
    ENDIF
ELSE
    OUTPUT "Minor"
ENDIF
```

What will be output?

- ○ **A.** Senior
- ○ **B.** Adult
- ○ **C.** Minor
- ○ **D.** No output

Q24 (3 marks)

A hospital is developing a patient management system. The development team must decide how to approach the design. Explain how decomposition and abstraction would each contribute differently to this project.

- ○ **A.** Both decomposition and abstraction mean the same thing in this context
- ○ **B.** Decomposition identifies system components (patient registration, appointment scheduling, medical records, billing, prescription management); abstraction hides complexity within each component (e.g., billing shows 'generate invoice' without exposing database queries, payment gateway integration, or accounting rules). Decomposition divides the project; abstraction manages complexity within divisions.
- ○ **C.** Only decomposition is relevant to hospital systems
- ○ **D.** Neither technique applies to medical software

Q25 (3 marks)

A flowchart shows the following algorithm:

```
Start → INPUT num → Set result to 1 → Is num > 1? → (Yes) result
```

If the user inputs 4, what will be output and what does this algorithm calculate?

- ○ **A.** Output: 4; the algorithm outputs the input unchanged
- ○ **B.** Output: 10; the algorithm adds numbers from 1 to n
- ○ **C.** Output: 24; the algorithm calculates the factorial of the input (n!)
- ○ **D.** Output: 16; the algorithm calculates the square of the input

Q26 (3 marks)

Consider the following pseudocode:

```
array data = [8, 3, 5, 1, 9, 2]
FOR i = 0 TO 4
    IF data[i] > data[i+1] THEN
        temp = data[i]
        data[i] = data[i+1]
        data[i+1] = temp
    ENDIF
ENDFOR
```

Analyse what this algorithm does and what the array will contain after one complete pass.

- ○ **A.** The array will be completely sorted: [1, 2, 3, 5, 8, 9]
- ○ **B.** The algorithm reverses the array: [2, 9, 1, 5, 3, 8]
- ○ **C.** This is one pass of bubble sort; it compares adjacent elements and swaps if out of order. After one pass: [3, 5, 1, 8, 2, 9] — the largest value (9) has 'bubbled' to the end
- ○ **D.** The algorithm finds the minimum value: output is 1

Q27 (3 marks)

A student creates a structure diagram for a 'Social Media App' with first-level decomposition showing: 'User Interface', 'Database', 'Server', 'Network Connection'.
Analyse whether this is good decomposition and explain your reasoning.

- ○ **A.** This is excellent decomposition that follows best practices
- ○ **B.** This decomposition is poor because it focuses on technical architecture layers rather than user-facing features/functions. Better decomposition would show: 'User Profiles', 'Posts/Feed', 'Messaging', 'Notifications', 'Search'. Functional decomposition reveals what the system does for users, not how it's technically implemented.
- ○ **C.** The decomposition has too few components
- ○ **D.** Social media apps cannot be decomposed

Q28 (3 marks)

A developer notices these similar code segments in their project:

- Check if username is between 3-20 characters
- Check if password is between 8-30 characters
- Check if postcode is between 5-8 characters

Using pattern recognition, design the most appropriate solution.

- ○ **A.** Keep all three as separate functions since they check different things
- ○ **B.** Create one function called 'checkUsername' and copy-paste it twice with different names
- ○ **C.** Create a generalised function 'validateLength(text, minLength, maxLength)' that accepts any string and any length range as parameters, returning true if the text length falls within the specified range. Call this function with appropriate parameters for each validation.

○ **D.** Remove all validation as it's unnecessary

Q29 (3 marks)

A company needs to develop a route-planning system for delivery drivers. Apply computational thinking principles to outline how you would approach this problem.

○ **A.** Start coding immediately to save time

○ **B.** Apply all four pillars: Decomposition breaks the problem into route calculation, traffic monitoring, delivery scheduling, and driver assignment. Abstraction simplifies the road network to nodes (locations) and edges (roads with distances/times), ignoring road surface details. Pattern recognition identifies that shortest-path algorithms used in mapping apps apply here. Algorithmic thinking designs the step-by-step process for finding optimal routes.

○ **C.** Hire more drivers instead of building software

○ **D.** Use pattern recognition only — the other pillars are unnecessary

Q30 (3 marks)

Two students write algorithms to find if a number exists in a sorted list of 1,000 items.
Student A: Check each item from the start until found or reach the end.
Student B: Check the middle item; if target is smaller, search the first half; if larger, search the second half; repeat.
Analyse the efficiency of both approaches.

○ **A.** Both approaches are equally efficient

○ **B.** Student A's approach is more efficient because it's simpler

○ **C.** Student A uses linear search (checking up to 1,000 items in worst case). Student B uses binary search (checking at most ~10 items,

since $\log_2 1000 \approx 10$). Binary search is dramatically more efficient for sorted data, reducing comparisons from potentially 1,000 to about 10.

○ **D.** Neither approach would work for finding numbers in a list

Total: **55** marks • Time: **30 min**

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